INDIVIDUAL CASE SAFETY REPORT (ICSR)

Task 5: Complete Pharmacovigilance Case Assessment Handbook

1. Introduction to ICSR Assessment

An Individual Case Safety Report (ICSR) is the primary document used in pharmacovigilance to record, analyze, and communicate single adverse event occurrences linked to medicinal products. Drug Safety Associates (DSAs) process ICSRs by evaluating clinical reports received from healthcare professionals, clinical trials, regulatory agencies, literature, and consumers.

Processing an ICSR requires executing four standardized operational deliverables: ADR Identification, ICSR Documentation, Causality Assessment, and Severity and Expectedness Classification. This comprehensive handbook provides a complete simulation of the ICSR workflow for clinical pharmacovigilance.

2. The Four Deliverables of a Drug Safety Associate

- Deliverable 01 - ADR Identification: Validating the four mandatory elements of a report and determining whether an adverse event is drug-related.
- Deliverable 02 - Reporting & ICSR Documentation: Structuring clinical narratives, mapping fields to CIOMS I formats, and coding terms using MedDRA.
- Deliverable 03 - Causality Assessment: Scoring drug-event relationships using validated Naranjo and WHO-UMC algorithms.
- Deliverable 04 - Severity & Case Classification: Evaluating ICH E2D seriousness criteria, expectedness against reference safety information, and regulatory submission deadlines.

3. Overview Matrix of ICSR Processing Workflow

Workflow Stage	Primary Tasks & Actions	Key Output / Deliverable
1. Triage & Intake	Verify presence of the four minimum valid ICSR elements.	Valid Case Identification
2. Data Entry & Coding	Enter patient history, suspect drugs, and code terms with MedDRA.	Structured ICSR / CIOMS Form
3. Causality Evaluation	Evaluate timeline, dechallenge, rechallenge, and alternative causes.	Naranjo & WHO Causality Category
4. Case Classification	Apply ICH E2D seriousness criteria and check SmPC expectedness.	SUSAR / Expeditable Status
5. Submission Triage	Determine regulatory submission timeframe (7-day or 15-day).	E2B Electronic Transmission

4. Deliverable 01: ADR Identification & Case Validation

Prior to conducting formal medical evaluations, a Drug Safety Associate must confirm that an incoming report meets global regulatory requirements for a valid Individual Case Safety Report. According to ICH E2D guidelines, a case is valid if it contains four mandatory elements.

4.1 The Four Minimum Criteria for a Valid ICSR

Mandatory Element	Regulatory Requirement	Acceptable Data Inputs
1. Identifiable Patient	At least one piece of information establishing patient existence.	Initials, age, gender, weight, patient ID, or date of birth.
2. Identifiable Reporter	Primary source contact details to permit follow-up.	Name, designation, clinic address, email, or telephone.
3. Suspect Medicinal Product	At least one pharmaceutical product linked to the event.	Brand name, generic name, active substance, or batch number.
4. Adverse Event / Reaction	At least one untoward medical outcome or symptom.	Clinical description, sign, lab anomaly, or disease term.

4.2 Initial Assessment of Case Validity

If any of the four mandatory elements are missing, the report is classified as invalid and cannot be transmitted to health authorities. In such instances, the Drug Safety Associate initiates a formal follow-up request to the primary reporter to gather the missing information.

4.3 Primary Master Case Study Presentation

PRIMARY MASTER CLINICAL CASE PROFILE (FOR TASK 5 ANALYSIS)	
Patient Demographics Female, 45 years old (Initials: B.M.), Weight: 64 kg	Primary Reporter Dr. A. Sharma, MD (Consultant Physician, City Clinic)
Suspect Medication Amoxicillin-Clavulanate 1000mg PO BID (Started 5 days ago)	Reported Adverse Event Acute facial edema, widespread urticarial rash, dyspnea, nausea

Validity Verdict: VALID CASE. All four mandatory elements (Identifiable Patient, Reporter, Suspect Drug, and Adverse Event) are present.

5. Deliverable 02: ICSR Documentation (CIOMS I Standard)

Below is the completed standardized Council for International Organizations of Medical Sciences (CIOMS I) form mapping for the Master Case Study.

I. REACTION INFORMATION

1. Patient Initials	1a. Age / Gender	2. Date of Reaction Onset	3. Reaction Outcome
B.M.	45 Yrs / Female	12/11/2026	Recovered

8-12. Description of Reaction

Patient developed localized facial swelling, generalized urticarial rash, moderate shortness of breath, and nausea approximately 50 minutes after taking the 9th dose of oral Amoxicillin-Clavulanate. Symptoms required urgent emergency room administration of intramuscular epinephrine and antihistamines.

II. SUSPECT DRUG(S) INFORMATION

14. Suspect Drug Name	15. Daily Dose & Route	16. Indication	17. Start Date	18. Stop Date
Amoxicillin-Clavulanate 1000mg	1000 mg Oral BID	Sinusitis	07/11/2026	12/11/2026

III. CONCOMITANT DRUGS & HISTORY

Concomitant Drug Name	Dose & Route	Start Date	Indication
Multivitamin Tablet	1 Tablet Oral Daily	01/01/2025	Dietary Supplement

IV. MANUFACTURER / REPORTER INFORMATION

24a. Reporter Name & Address	24b. Date of Report
Dr. A. Sharma, MD, City Clinic	13/11/2026

6. MedDRA Terminology Coding & Hierarchy

Adverse reaction text reported in raw clinical narratives must be converted into standard medical terms using the Medical Dictionary for Regulatory Activities (MedDRA) hierarchy to facilitate global database retrieval.

6.1 Five-Level MedDRA Hierarchy Breakdown

MedDRA organizes terms into five structured levels, moving from broad physiological systems down to exact reported phrases:

- System Organ Class (SOC): Highest level of anatomical or physiological grouping.
- High-Level Group Term (HLGT): Broad sub-grouping of anatomical or pathological terms.
- High-Level Term (HLT): Specific disease or pathological process grouping.
- Preferred Term (PT): Single discrete medical concept used for official regulatory reporting.
- Lowest Level Term (LLT): Synonym or verbatim phrase as reported by the physician or patient.

6.2 MedDRA Coding Matrix for Master Case Study

Reported Symptom	System Organ Class (SOC)	Preferred Term (PT)	Lowest Level Term (LLT)
Facial Swelling	Skin and subcutaneous tissue disorders	Angioedema	Swelling of face
Urticarial Rash	Skin and subcutaneous tissue disorders	Urticaria	Hives / Allergic rash
Shortness of Breath	Respiratory, thoracic and mediastinal disorders	Dyspnea	Breathing difficulty
Nausea	Gastrointestinal disorders	Nausea	Feeling sick to stomach
Anaphylactic Reaction	Immune system disorders	Anaphylactic reaction	Acute penicillin anaphylaxis

6.3 Standardization of Case Narratives

A Drug Safety Associate synthesizes raw reporter accounts into a chronological, objective medical narrative. Standardized narratives detail baseline patient history, suspect drug administration, latency, clinical presentation, emergency interventions, lab diagnostics, dechallenge outcome, and final disposition without incorporating speculative opinions.

7. Deliverable 03: Causality Assessment Execution

Evaluating causality involves systematically applying established algorithms to determine whether evidence supports a drug-reaction link in the primary case study.

7.1 Naranjo Algorithm Evaluation for Master Case Study

Naranjo Assessment Parameter	Response	Points
1. Conclusive previous reports on this reaction?	Yes	+1
2. Event appeared after suspect drug was given?	Yes	+2
3. Reaction improved when drug was discontinued?	Yes	+1
4. Reaction reappeared on rechallenge?	Unknown	0
5. Alternative non-drug causes ruled out?	Yes	+2
6. Reaction reappeared with placebo?	Unknown	0
7. Toxic drug concentrations detected in blood?	Unknown	0
8. Reaction dose-dependent?	Unknown	0
9. Similar past reaction to penicillins?	Unknown	0
10. Objective clinical confirmation (ER record)?	Yes	+1
Total Naranjo Probability Score		**7 Points**

7.2 WHO-UMC Causality Classification

WHO-UMC QUALITATIVE EVALUATION SUMMARY	
Assigned Category PROBABLE / LIKELY	Clinical Rationale Event demonstrates a clear, plausible temporal relationship (onset 50 minutes post-dose) with positive dechallenge upon drug cessation. Symptoms cannot be reasonably attributed to underlying sinusitis or concomitant multivitamin use.

7.3 Pharmacological & Mechanism Summary

Amoxicillin forms penicilloyl hapten determinants that bind serum proteins, triggering an IgE-mediated Type I hypersensitivity cascade. Degranulation of tissue mast cells and circulation basophils releases histamine and leukotrienes, accounting for acute angioedema, urticaria, and dyspnea.

8. Deliverable 04: Severity & Expectedness Classification

The final operational deliverable requires evaluating regulatory seriousness criteria, analyzing reference safety information to determine expectedness, and establishing reporting timelines.

8.1 Statutory Seriousness Evaluation (ICH E2D Criteria)

Seriousness Criterion	Criterion Met?	Clinical Justification
1. Results in death?	No	Patient survived following intervention.
2. Life-threatening?	Yes	Acute dyspnea and airway angioedema.
3. Requires inpatient hospitalization?	No	Treated in emergency department.
4. Persistent / significant disability?	No	Full recovery without sequelae.
5. Congenital anomaly / Birth defect?	No	Not applicable.
6. Medically important event?	Yes	Required emergency IM epinephrine.
Final Seriousness Classification		**SERIOUS**

8.2 Expectedness Evaluation Against Approved SmPC

REFERENCE SAFETY INFORMATION (SMPC) COMPARISON		
Reference Document Amoxicillin SmPC Section 4.8	Expectedness Result EXPECTED	Expedited Status 15-Day Expedited ICSR
Evaluation Rationale The official Summary of Product Characteristics for Amoxicillin-Clavulanate explicitly lists acute urticaria, angioedema, and anaphylactic reactions under immune system disorders. Because the reaction is fully described in the reference safety document, it is classified as EXPECTED.		

8.3 Summary of Final ICSR Triage Classification

- Case Validity: Valid ICSR (Contains all 4 mandatory elements).
- Seriousness Rating: Serious (Life-threatening / Medically Important Event).
- Expectedness Rating: Expected (Listed in Section 4.8 of SmPC).
- Regulatory Action: Expedited ICSR transmission within 15 calendar days to regulatory authorities.

9. Secondary Case Assessment 2 & Case Assessment 3

To demonstrate broad clinical competency as a Drug Safety Associate, two additional secondary ICSR case scenarios are evaluated across all four deliverables.

9.1 Secondary Case 2: Ciprofloxacin Tendinitis

CASE 2 CLINICAL PROFILE & EVALUATION		
Patient Profile 68yo Male (Initials: H.T.)	Suspected Drug Ciprofloxacin 500mg PO	Adverse Event Achilles Tendinitis & Pain
Four Deliverables Summary 1. ADR Identification: Valid case (Patient H.T., Dr. Miller, Ciprofloxacin, Tendinitis). 2. MedDRA Coding: Tendinitis (PT), Achilles tendon pain (LLT), Musculoskeletal disorders (SOC). 3. Causality: Naranjo Score = 6 (Probable); WHO-UMC = Probable. 4. Classification: Non-Serious (No rupture/hospitalization), Expected (Boxed warning on SmPC). Periodic report.		

9.2 Secondary Case 3: Olanzapine-Induced DKA (SUSAR)

CASE 3 CLINICAL PROFILE & EVALUATION		
Patient Profile 28yo Female (Initials: C.R.)	Suspected Drug Olanzapine 10mg PO	Adverse Event Diabetic Ketoacidosis (DKA)
Four Deliverables Summary 1. ADR Identification: Valid case (Patient C.R., Dr. Evans, Olanzapine, DKA). 2. MedDRA Coding: Diabetic ketoacidosis (PT), Acute metabolic acidosis (LLT), Endocrine disorders (SOC). 3. Causality: Naranjo Score = 7 (Probable); WHO-UMC = Probable. 4. Classification: SERIOUS (ICU admission) and UNEXPECTED (Sudden DKA in non-diabetic). Classified as a SUSAR requiring 15-day expedited reporting.		

10. Electronic Transmission Standards (ICH E2B Guidelines)

Once an ICSR is documented and classified, Drug Safety Associates format the data into standardized XML structures defined by ICH E2B guidelines for electronic transmission to regulatory databases.

10.1 Key ICH E2B(R3) Data Field Modules

E2B(R3) Module	Module Description & Content	Database System Impact
Module A: Administrative	Case safety report unique identifier, sender details, receiver details, and report type.	Establishes global ICSR record key and origin tracing.
Module B.1: Patient	Demographics, age, gender, medical history, baseline characteristics, and laboratory tests.	Facilitates patient subgroup risk profiling.
Module B.2: Reaction	MedDRA terms, onset date, stop date, reaction duration, outcome, and seriousness criteria.	Feeds signal detection algorithms (PRR / ROR).
Module B.4: Drug(s)	Suspect/concomitant drug details, dosage, route, indication, start/stop dates, and dechallenge.	Enables drug exposure timeline reconstruction.
Module B.5: Narrative	Comprehensive clinical narrative summarizing case history, hospital findings, and assessment.	Supports medical review and expert audit.

10.2 Global Pharmacovigilance Regulatory Gateways

Completed E2B XML files are transmitted directly to national authority safety gateways including:

- FDA FAERS Gateway: United States Food and Drug Administration electronic portal.
- EudraVigilance Gateway: European Medicines Agency centralized safety database.
- WHO VigiBase: World Health Organization global ICSR repository managed by the Uppsala Monitoring Centre.

11. Quality Assurance & Post-Market Signal Detection

Individual case safety reports serve as the foundational data points used by pharmacovigilance epidemiologists to detect emerging safety signals across large healthcare populations.

11.1 Quality Review Checklist for Drug Safety Associates

Before locking and submitting an ICSR, a mandatory quality check is performed:

- Duplicate Check: Verify that the case has not been previously submitted under a different source identifier.
- Verbatim Verification: Ensure MedDRA Preferred Terms accurately reflect reported clinical symptoms without under-coding.
- Chronological Consistency: Confirm that reaction onset dates occur after drug start dates.
- Seriousness Alignment: Check that selected seriousness criteria match narrative documentation (e.g., hospitalization box checked only if inpatient stay occurred).
- Causality Logic: Confirm that Naranjo point entries align with written clinical justifications.

11.2 Disproportionality Analysis in Signal Detection

Statistical Metric	Mathematical Concept	Signal Threshold Criterion
Proportional Reporting Ratio (PRR)	Compares proportion of specific reaction for suspect drug versus all other drugs in database.	PRR ≥ 2.0 with Chi-Square ≥ 4.0 (N ≥ 3 cases)
Reporting Odds Ratio (ROR)	Odds of reporting specific reaction with suspect drug relative to odds for other drugs.	Lower 95% Confidence Interval > 1.0

12. Master Summary Table of Evaluated Task Cases

Case	Suspect Drug	MedDRA PT Term	Causality	Seriousness	Submission
Master 1	Amoxicillin-Clav	Angioedema / Urticaria	Probable (7)	Serious	15-Day ICSR
Case 2	Ciprofloxacin	Tendinitis	Probable (6)	Non-Serious	Periodic (PSUR)
Case 3	Olanzapine	Diabetic ketoacidosis	Probable (7)	Serious (SUSAR)	15-Day SUSAR

13. Standardized DSA Operational Checklist

ICSR PROCESSING DESK REFERENCE CHECKLIST

1. Validate Four Mandatory Elements: Confirm patient, reporter, drug, and event.
2. Perform MedDRA Medical Coding: Select accurate Preferred Terms (PT) for all reported signs.
3. Conduct Causality Assessment: Execute Naranjo algorithm and assign WHO-UMC category.
4. Classify Seriousness & Expectedness: Apply ICH E2D criteria and check SmPC labeling.
5. Transmit via E2B Gateway: Format XML and submit within mandated 7-day or 15-day window.

14. Final Conclusion

The Individual Case Safety Report (ICSR) workflow forms the core operational pillar of post-marketing drug safety. By executing the four key deliverables—ADR identification, CIOMS/MedDRA documentation, causality scoring, and severity classification—Drug Safety Associates ensure regulatory compliance, support early safety signal detection, and protect global public health.